

- 12 (a) e.g. signal can be regenerated
 so that there is minimal noise
 e.g. extra data can be added M1
 so that signal can be checked for errors A1
 (any two, sensible suggestions, M1 + A1, max 4)
- (b) (i) 1101 B1 [1]
 (ii) 5 B1 [1]
- (c) (i) block X: serial-to parallel B1
 block Y: DAC / digital-to-analogue (converter) B1 [2]
 (ii) takes the simultaneous / all bits of a number M1
 and transmits them one after another / down a single line A1 [2]
- (d) increase number of bits in digital number at each sampling M1
 so that step height is reduced A1
 increase sampling frequency / reduce time between samples M1
 so that depth / width of step is reduced A1 [4]
 (do not allow 'smoother output')

[Total: 14]